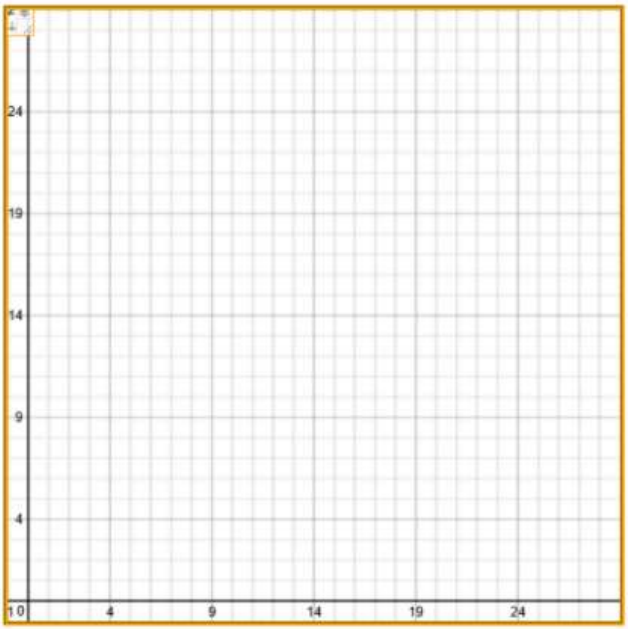
1) Find the maximum value of the objective function $z = x + 3y$

subject to the following constraints: $\begin{cases} x \geq 0 \text{ and } y \geq 0 \\ x + 3y \leq 6 \\ x - y \leq 4 \end{cases}$



a) When $x = 0$ and $y = 0$, $z =$ __?__

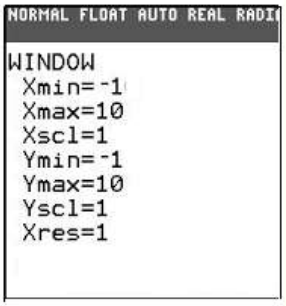
When $x = 0.5$ and $y = 0.5$, $z =$ __?__

When $x = 1$ and $y = 1$, $z =$ __?__

When $x = 1$ and $y = 1.5$, $z =$ __?__

When $x = 2$ and $y = 2$, $z =$ __?__

When $x = 2$ and $y = 1.5$, $z =$ __?__



b) Solve the equation $x + 3y = 6$ for y. $y =$ __?__

c) Solve the equation $x - y = 4$ for y. $y =$ __?__

d) Find the intersection of the lines $x + 3y = 6$ and $x - y = 4$.
Intersection = __?__

e) Draw the region representing the constraints.

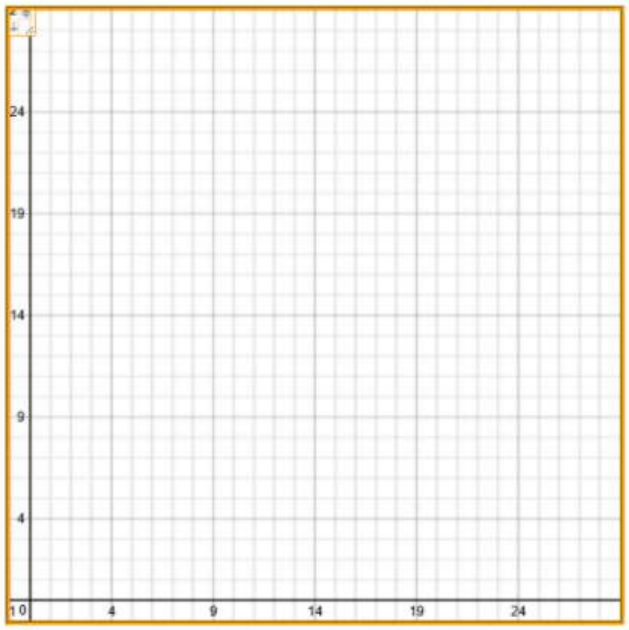
f) Find y-intercept of the line $x + 3y = 6$. Hint: Set $x = 0$.

g) Find x-intercept of the line $x - y = 4$. Hint: Set $y = 0$.

h) Find the maximum value of the objective function $z = x + 3y$. $x =$ __?__ and $y =$ __?__

2) Find the maximum value of the objective function $z = 4x + 2y$

subject to the following constraints: $\begin{cases} x \geq 0 \text{ and } y \geq 0 \\ 2x + 3y \leq 6 \\ 2x - y \leq 4 \end{cases}$



a) When $x = 0$ and $y = 0$, $z =$ __?__

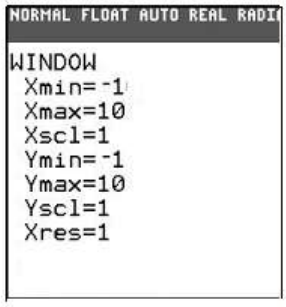
When $x = 0.5$ and $y = 0.5$, $z =$ __?__

When $x = 1$ and $y = 1$, $z =$ __?__

When $x = 1$ and $y = 1.5$, $z =$ __?__

When $x = 2$ and $y = 2$, $z =$ __?__

When $x = 2$ and $y = 1.5$, $z =$ __?__



b) Solve the equation $2x + 3y = 6$ for y. $y =$ __?__

c) Solve the equation $2x - y = 4$ for y. $y =$ __?__

d) Find the intersection of the lines $2x + 3y = 6$ and $2x - y = 4$.
Intersection = __?__

e) Draw the region representing the constraints.

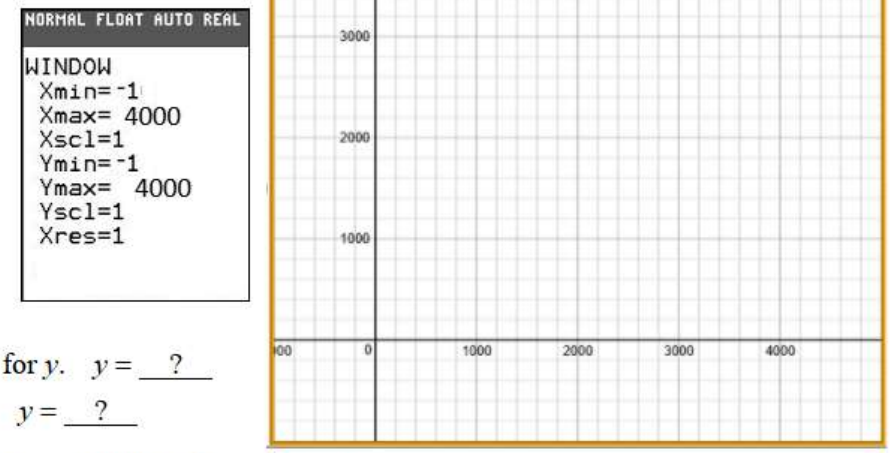
f) Find y-intercept of the line $2x + 3y = 6$. Hint: Set $x = 0$.

g) Find x-intercept of the line $2x - y = 4$. Hint: Set $y = 0$.

h) Find the maximum value of the objective function $z = 4x + 2y$. $x =$ __?__ and $y =$ __?__

3) Find the maximum value of the objective function $z = 5x + 3y$

subject to the following constraints: $\begin{cases} x \geq 0 \text{ and } y \geq 0 \\ 30x + 15y \leq 60,000 \\ x + y \leq 3,000 \end{cases}$



a) When $x = 0$ and $y = 0$, $z =$ __?__

When $x = 0.5$ and $y = 0.5$, $z =$ __?__

When $x = 1$ and $y = 1$, $z =$ __?__

When $x = 1$ and $y = 1.5$, $z =$ __?__

When $x = 2$ and $y = 2$, $z =$ __?__

When $x = 2$ and $y = 1.5$, $z =$ __?__

b) Solve the equation $30x + 15y = 60,000$ for y. $y =$ __?__

c) Solve the equation $x + y = 3,000$ for y. $y =$ __?__

d) Find the intersection of the lines $30x + 15y = 60,000$ and $x + y = 3,000$.

Intersection = __?__

e) Draw the region representing the constraints.

f) Find x-intercept of the line $30x + 15y = 60000$. Hint: Set $y = 0$.

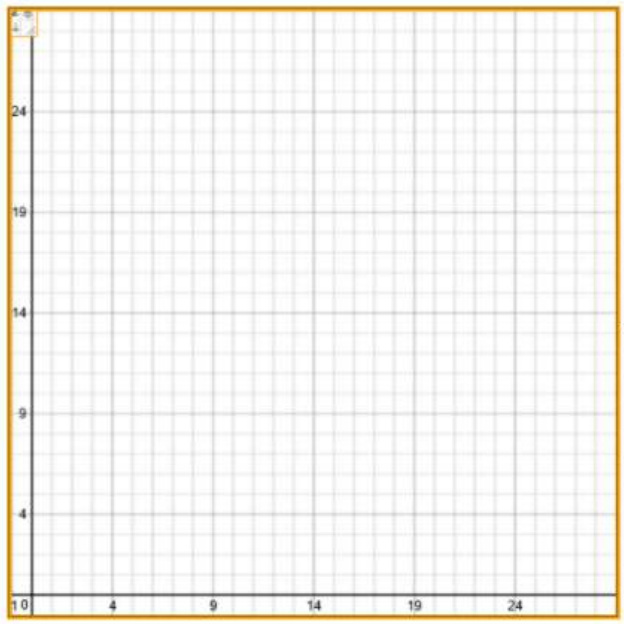
g) Find y-intercept of the line $x + y = 3000$. Hint: Set $x = 0$.

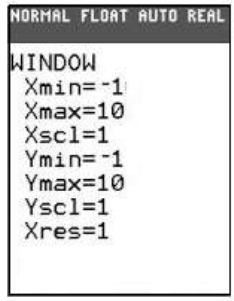
h) Find the maximum value of the objective function $z = 5x + 3y$. $x =$ __?__ and $y =$ __?__

4) Suppose the profit function is $P = 4x + 12y$

Find the maximum value of the objective function $P = 4x + 12y$

subject to the following constraints: $\begin{cases} x \geq 0 \text{ and } y \geq 0 \\ 2x + y \leq 18 \\ 2x + 4y \leq 32 \end{cases}$





a) When $x = 0$ and $y = 0$, $z =$ __?__

When $x = 0.5$ and $y = 0.5$, $z =$ __?__

When $x = 1$ and $y = 1$, $z =$ __?__

When $x = 1$ and $y = 1.5$, $z =$ __?__

When $x = 2$ and $y = 2$, $z =$ __?__

When $x = 2$ and $y = 1.5$, $z =$ __?__

b) Solve the equation $2x + y = 18$ for y. $y =$ __?__

c) Solve the equation $2x + 4y = 29$ for y. $y =$ __?__

d) Find the intersection of the lines $2x + y = 18$ and $2x + 4y = 29$.

Intersection = __?__

e) Draw the region representing the constraints.

f) Find x-intercept of the line $2x + y = 18$. Hint: Set $y = 0$.

g) Find y-intercept of the line $2x + 4y = 29$. Hint: Set $x = 0$.

h) Find the maximum value of the objective function $P = 4x + 12y$. $x =$ __?__ and $y =$ __?__